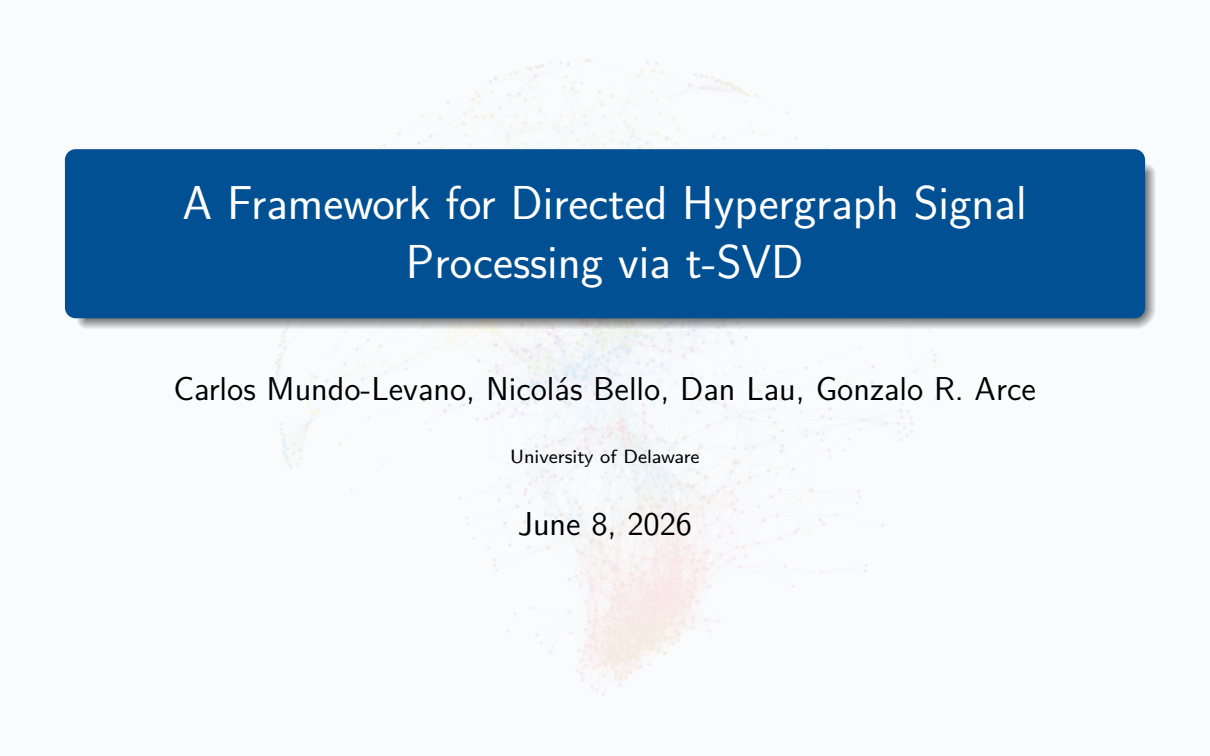




A Framework for Directed Hypergraph Signal Processing via t-SVD

Carlos Mundo-Levano, Nicolás Bello, Dan Lau, Gonzalo R. Arce

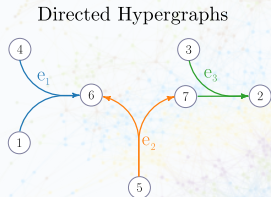
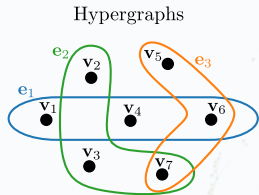
University of Delaware

June 8, 2026

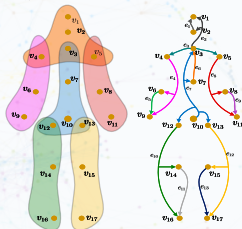
Outline

- ① Motivation
- ② Directed Hypergraph Algebraic Descriptor
- ③ Background
- ④ Shift Operator
- ⑤ Directed Hypergraph Fourier Transform
- ⑥ Experimental Results
- ⑦ Conclusion

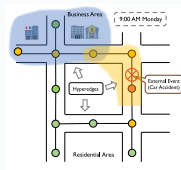
Motivation



Skeleton Based Human Pose



Traffic Forecasting



Hypergraphs

- Hyperedge = node subset
- Symmetric, undirected
- Group (polyadic) relations

Group flows, source $\rightarrow$ target

Directed Hypergraphs

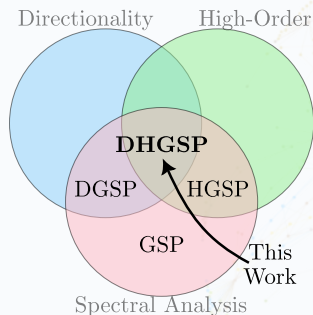
- Hyperarc: Tail $\rightarrow$ Head
- Asymmetric, directional
- B-hyperarc: single head node

Directed & higher-order across domains:

- Chemical reaction networks
- Metabolic pathways
- Gene regulatory networks
- Protein-signaling cascades
- Recommendation systems
- Information diffusion
- Citation / knowledge graphs
- Supply chains & logistics

Skeleton pose: Cui et al. (2024). Traffic forecasting: Zhao et al. (2023, 2024).

Motivation



- **GSP**: Undirected, pairwise only
- **DGSP** (directed-graph SP): Directed, no canonical FT
- **t-HGSP**: High-order, not directed

Challenge

No canonical DHG encoding

No canonical FT for DHGs

No spectral framework for DHGs

Contribution

B-hyperarc (single head): canonical tensor, shift & FT

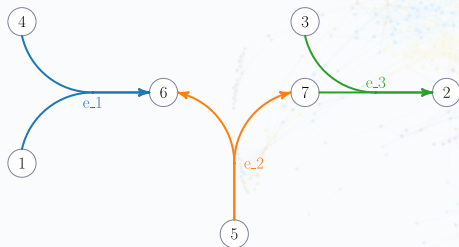
Lossless **t-DHGFT** via t-SVD

Outperforms all baselines

Key: $\text{GSP}, \text{DGSP}, \text{t-HGSP} \subsetneq \text{t-DHGSP}$

Directed Hypergraph

DH = (V, E): hyperarc $e = (T(e), H(e))$ — tail (sources) $\rightarrow$ head (targets).



$$V = \{v_1, \dots, v_7\}$$

$$E = \{\{v_1, v_4\} \rightarrow \{v_6\}, \{v_3, v_7\} \rightarrow \{v_2\}, \{v_5\} \rightarrow \{v_6, v_7\}\}$$

Why existing representations fail:

No canonical head index $\Rightarrow$ **identifiability failure**

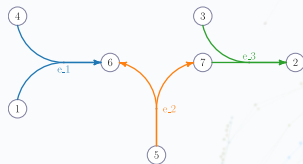
Shift mixes distinct head signals $\Rightarrow$ **cross-talk**

Challenges:

- No algebraic structure for DHGs
- No canonical FT for DHGs
- No unified spectral framework

B-hyperarc: $|H(e)| = 1$ — unique head encoding, eliminates both problems

B-Hyperarcs: Canonical Decomposition



General directed hypergraph

$$e_2 = \{v_5\} \rightarrow \{v_6, v_7\}$$

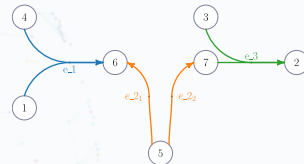
splits into:

$$e_{2_1} = \{v_5\} \rightarrow \{v_6\}$$

$$e_{2_2} = \{v_5\} \rightarrow \{v_7\}$$

Lossless: $T(e_i)$ fully preserved
in each e_{i_k} ; union covers $H(e_i)$.

No tail-to-head relationship
dropped.

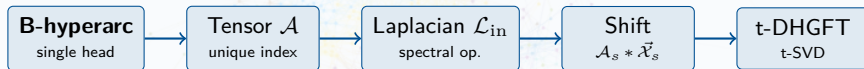


B-hypergraph (one head each)

First index = head node. Unique encoding: no identifiability failure, no cross-talk.

All tail→head connections preserved. Co-targeting has no algebraic effect on shift or FT in this framework.

B-Hyperarc: The Algebraic Foundation



One head per hyperarc. Everything else follows naturally.

Adjacency Tensor $\mathcal{A} \in \mathbb{R}^{N^M}$, M -th order

M = max hyperarc size; $\mathcal{A}$ has M indices, first = head.

For B-hyperarc $e_i = (T(e_i), \{v_k\})$, fill all $(M-1)$ -permutations of $T(e_i)$:

$$a_{k, p_2, \dots, p_M} = \frac{|T(e_i)|}{\alpha_i}$$

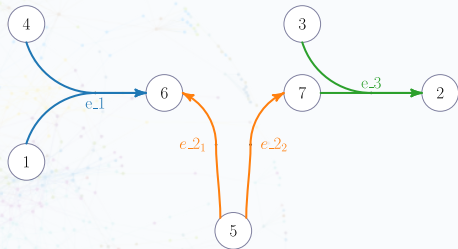
α_i : # ordered $(M-1)$ -permutations of $T(e_i)$

In-degree tensor $\mathcal{D}_{\text{in}} \in \mathbb{R}^{N^M}$:

fills **superdiagonal** only:

$$d_{\underbrace{k, k, \dots, k}_M} = \sum_{i_2, \dots, i_M} a_{k, i_2, \dots, i_M}$$

$$\mathcal{L}_{\text{in}} = \mathcal{D}_{\text{in}} - \mathcal{A}$$



$M = 3$: $\mathcal{A} \in \mathbb{R}^{N \times N \times N}$, indices (i_1, i_2, i_3) , $i_1 = \text{head}$

$$e_1 : \{v_1, v_4\} \rightarrow \{v_6\}: a_{6,1,4} = a_{6,4,1} = 1$$

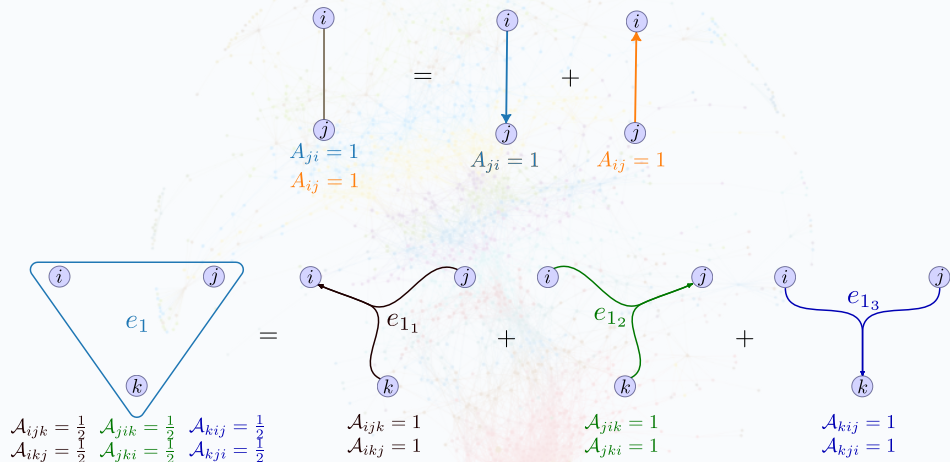
$$e_{2_1} : \{v_5\} \rightarrow \{v_6\}: a_{6,5,5} = 1$$

$$e_{2_2} : \{v_5\} \rightarrow \{v_7\}: a_{7,5,5} = 1$$

$$e_3 : \{v_3, v_7\} \rightarrow \{v_2\}: a_{2,3,7} = a_{2,7,3} = 1$$

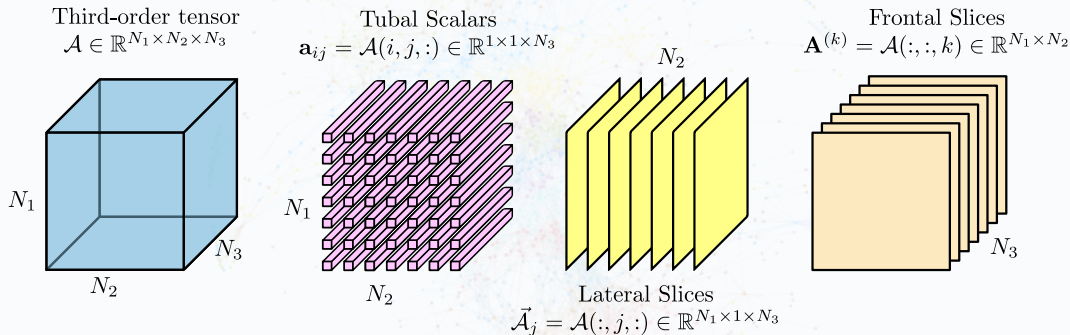
In-degree v_6 : $d_{6,6,6} = a_{6,1,4} + a_{6,4,1} + a_{6,5,5} = 3$

Directed Hypergraphs: A Unifying Structure



Undirected edge $\Rightarrow$ pair of B-hyperarcs. Graph ($M = 2$) and undirected hypergraph ($M > 2$) are special cases.

Tensor Notation



Kilmer, M., et al. "Third-order tensors as operators on matrices." (2013).

Symmetrization Preserves Directionality

Symmetrization $\text{sym}(\cdot)$ reflects along modes $p \geq 3$:

$$\mathcal{A}_s = \text{sym}(\mathcal{A}) = \frac{1}{2} [\mathbf{0} \mid \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \mid \mathbf{A}^{(N)}, \dots, \mathbf{A}^{(1)}], \quad \mathcal{A}_s \in \mathbb{R}^{N \times N \times N_s^{M-2}}$$

$N_s = 2N + 1$; frontal slices stacked along mode 3. **Shift**: first frontal slice of output preserves spatial structure. **t-SVD**: palindromic mode-3 $\Rightarrow$ real DFT slices $\Rightarrow$ real singular values.

- Reflection touches modes ≥ 3 only — never the frontal slice
- Each slice $\mathbf{A}^{(k)}$: rows = heads, columns = tails
- Asymmetric: $\mathbf{A}^{(k)} \neq (\mathbf{A}^{(k)})^\top$ when head $\neq$ tail

Cyclic symmetry in modes $\geq 3 \Rightarrow$ DFT-based computation; **frontal slices untouched**
 $\Rightarrow$ **directionality intact.**

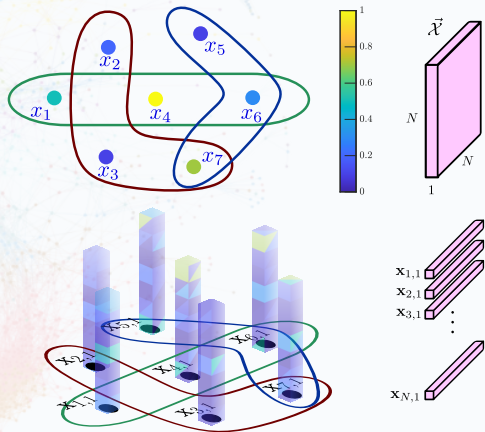
Signals on Hypergraphs

Notation: $\vec{\mathcal{X}}$ — signal tensor (arrow: vector of tubal scalars, analogous to a signal vector in standard GSP). $(\cdot)_s$: symmetrized form (previous slide).

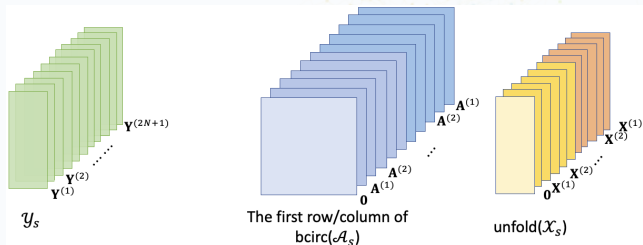
- A hypergraph signal is an N -length vector of tubal scalars $\vec{\mathcal{X}} \in \mathbb{R}^{N \times 1 \times N}$.
- Each tubal scalar $\mathbf{x}_{i,1}$ in $\vec{\mathcal{X}}$ is obtained from a one dimensional signal in the hypergraph $\mathbf{x} \in \mathbb{R}^N$ as

$$\mathbf{x}_{i,1} = \text{fold} \left(\begin{bmatrix} x_i x_1 \\ x_i x_2 \\ \vdots \\ x_i x_N \end{bmatrix} \right).$$

- Or $\vec{\mathcal{X}} = x \odot x$ ($\odot$: outer product) and $\vec{\mathcal{X}}_s = \text{sym}(\vec{\mathcal{X}})$



t-Product / Shift Operator



$$\begin{aligned}\mathcal{A}_s &\in \mathbb{R}^{N \times N \times (2N+1)} \\ \mathcal{X}_s &\in \mathbb{R}^{N \times D \times (2N+1)} \\ \mathcal{Y}_s &\in \mathbb{R}^{N \times D \times (2N+1)} \\ D: &\text{signal columns}\end{aligned}$$

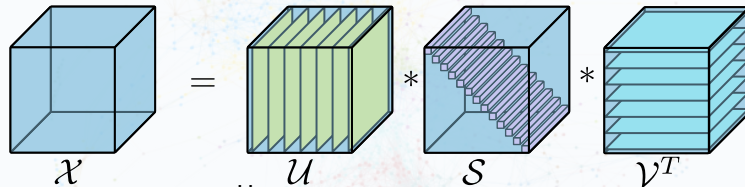
$$\mathcal{Y}_s = \mathcal{A}_s * \mathcal{X}_s = \frac{1}{2} \text{fold}(\text{bcirc}(\mathcal{A}_s) \cdot \text{unfold}(\mathcal{X}_s))$$

$$= \frac{1}{2} \text{fold} \left(\begin{bmatrix} \mathbf{0} & \mathbf{A}^{(1)} & \mathbf{A}^{(2)} & \dots & \mathbf{A}^{(2N)} & \mathbf{A}^{(2N+1)} \\ \mathbf{A}^{(1)} & \mathbf{0} & \mathbf{A}^{(1)} & \dots & \mathbf{A}^{(2N)} & \mathbf{A}^{(2N+1)} \\ \mathbf{A}^{(2)} & \mathbf{A}^{(1)} & \mathbf{0} & \dots & \mathbf{A}^{(2N-1)} & \mathbf{A}^{(2N)} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{A}^{(2N)} & \mathbf{A}^{(2N-1)} & \mathbf{A}^{(2N-2)} & \dots & \mathbf{0} & \mathbf{A}^{(1)} \\ \mathbf{A}^{(2N+1)} & \mathbf{A}^{(2N)} & \mathbf{A}^{(2N-1)} & \dots & \mathbf{A}^{(1)} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \\ \vdots \\ \mathbf{X}^{(2N)} \\ \mathbf{X}^{(2N+1)} \end{bmatrix} \right)$$

t-HGSP [Pena-Pena, K., Lau, D.L., Arce, G.R. "T-HGSP: Hypergraph signal processing using t-product tensor decompositions." IEEE Trans. Signal Inf. Process. Netw., 9:329–345, 2023]: signal processing framework for hypergraphs using t-product and t-eigendecomposition.

Tensor SVD (t-SVD)

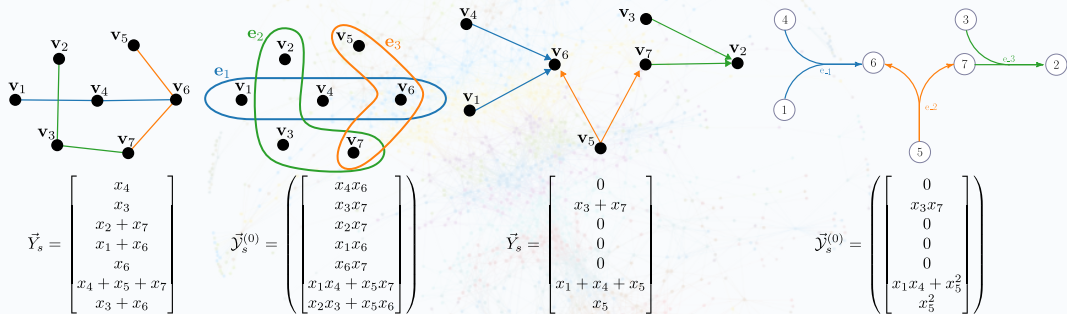
For $\mathcal{L} \in \mathbb{R}^{N \times N \times N_s}$:

$$\mathcal{L} = \mathcal{U} * \mathcal{S} * \mathcal{V}^H$$


- $\mathcal{U}, \mathcal{V}$: t-orthogonal ($\mathcal{U}^H * \mathcal{U} = \mathcal{I}$)
- $\mathcal{S}$: f-diagonal, tubal singular values $\sigma_1 \leq \dots \leq \sigma_N$
- Computed via DFT: one matrix SVD per frequency slice

Kilmer, M., et al. "Factorization strategies for third-order tensors." Lin. Alg. Appl. (2011).

Shift Operation: GSP vs HGSP vs DHGSP



GSP: scalar diffusion on edges. t-HGSP: tensor diffusion on undirected hyperarcs. t-DHGSP: tensor diffusion respecting tail $\rightarrow$ head direction.

Topological Localization

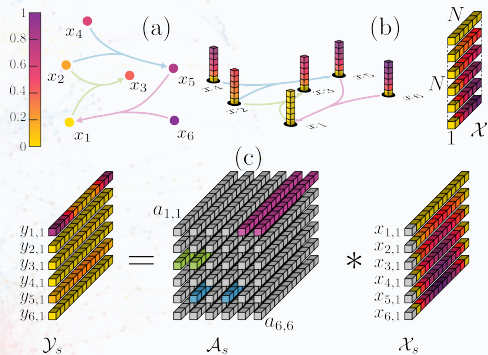
Theorem

Let $\vec{\mathcal{X}}$ be localized at v ($x_v = 1$, $x_u = 0$ otherwise).

$\mathcal{Y}_s^{(\ell)} = \mathcal{A}_s^\ell * \mathcal{X}_s$ is nonzero only at nodes reachable from v in exactly ℓ directed hyperarc steps.

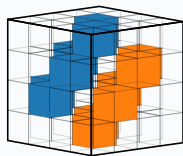
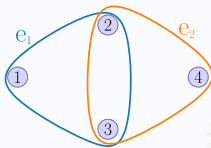
$\mathcal{A}_s^\ell$: $\mathcal{A}_s$ t-product-multiplied with itself ℓ times.

Consequence: shift respects directed topology, prerequisite for causal spectral analysis.

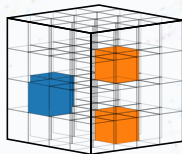
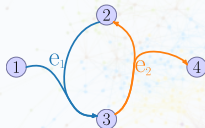


Challenge: Defining a Fourier Transform for Directed Hypergraphs

Undirected Hypergraph



Directed Hypergraph



Frontal Slices are not symmetrical

not exist $\iff$ no Lossless Fourier Transform

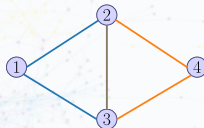
Full Eigen-Decomposition could

be no orthonormal $\iff$ no Parseval's Identity

Frontal slices of $\mathcal{L}_{in}$ are asymmetric (directed) — t-SVD alone yields complex eigenbasis. Need:

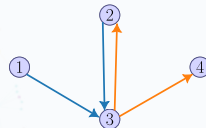
self-adjoint dilation.

Graph



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Directed Graph



$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Matrix Representation
not symmetric

Self-Adjoint Dilation

Built on B-hyperarc in-Laplacian $\mathcal{L}_{\text{in}} = \mathcal{D}_{\text{in}} - \mathcal{A}$ Self-adjoint dilation of $\mathcal{L}_{\text{in}_s} \in \mathbb{R}^{N \times N \times N_s^{M-2}}$:

$$\begin{pmatrix} \mathbf{O} & \mathcal{L}_{\text{in}_s} \\ \mathcal{L}_{\text{in}_s}^T & \mathbf{O} \end{pmatrix} \in \mathbb{R}^{2N \times 2N \times N_s^{M-2}}$$

- Symmetric by construction — real orthonormal eigenbasis
- t-SVD: $\mathcal{L}_{\text{in}_s} = \mathcal{U} * \mathcal{S} * \mathcal{V}^H$
- Eigenbasis of dilation: $\mathcal{W} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathcal{U} & \mathcal{U} \\ \mathcal{V} & -\mathcal{V} \end{pmatrix}$

Chen, Y., Cheng, C., Sun, Q. "Graph Fourier transform based on singular value decomposition of the directed Laplacian." *Sampl. Theory Signal Process. Data Anal.*, 21 (2023). DOI:10.1007/s43670-023-00062-w

t-DHGFT and Inverse

Embed $\mathcal{X}_s$ isometrically: $\mathcal{X}_s = \frac{1}{\sqrt{2}}(\mathcal{I}_s; \mathcal{I}_s)^\top * \mathcal{X}_s$ ($\mathcal{I}_s$: t-product identity; $\mathcal{X}_s$: signal in $2N$ -dim dilation space)

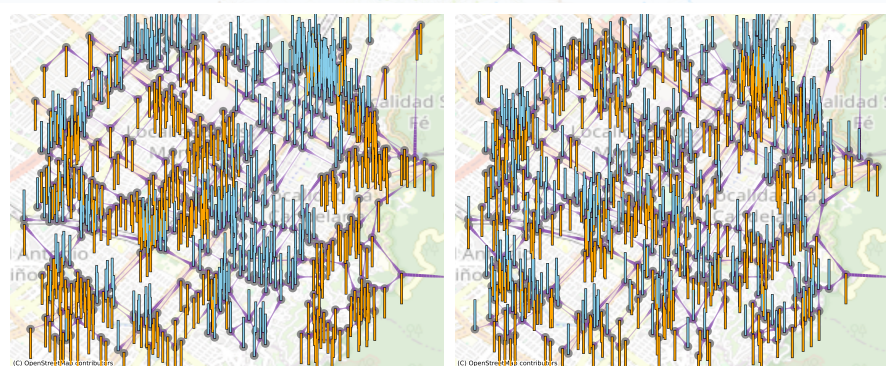
$$\vec{\mathcal{X}}_{F_s} = \mathcal{W}^H * \mathcal{X}_s = \frac{1}{2} \begin{pmatrix} (\mathcal{U}^\top + \mathcal{V}^\top) * \mathcal{X}_s \\ (\mathcal{U}^\top - \mathcal{V}^\top) * \mathcal{X}_s \end{pmatrix}$$

$$\vec{\mathcal{X}}_s = \frac{1}{2} \left(\mathcal{U} * (\vec{\mathcal{X}}_{F_s,1} + \vec{\mathcal{X}}_{F_s,2}) + \mathcal{V} * (\vec{\mathcal{X}}_{F_s,1} - \vec{\mathcal{X}}_{F_s,2}) \right)$$

$\vec{\mathcal{X}}_{F_s}$: Fourier-domain signal; subscripts 1,2 index upper/lower halves of the $2N$ -dim dilation.

- $\mathcal{W}^H * \mathcal{W} = \mathcal{I}$ — perfect reconstruction
- Parseval's identity guaranteed
- Frequencies: $\sigma_1 \leq \sigma_2 \leq \dots$ (ascending singular values)

Frequency Ordering — Real Traffic Network



First frontal slices of $\hat{\mathbf{V}}$ (right singular vectors of $\mathcal{L}_{in_s}$ t-SVD).

Left: low-freq ($\sim 5\%$) — spatially smooth.

Right: high-freq ($\sim 95\%$) — rapid oscillation.

Bogotá, Colombia traffic network (689 nodes). GSP-Traffic dataset [Zhao et al., 2024].

Denoising Setup

Dataset

- Macheng, China traffic (153 nodes)
- B-hyperarcs: co-arriving flows grouped $(v_1 \rightarrow v_3, v_2 \rightarrow v_3 \Rightarrow \{v_1, v_2\} \rightarrow \{v_3\})$

Method

- AWGN corruption, 100 trials
- Low-pass hard threshold in t-DHGFT domain

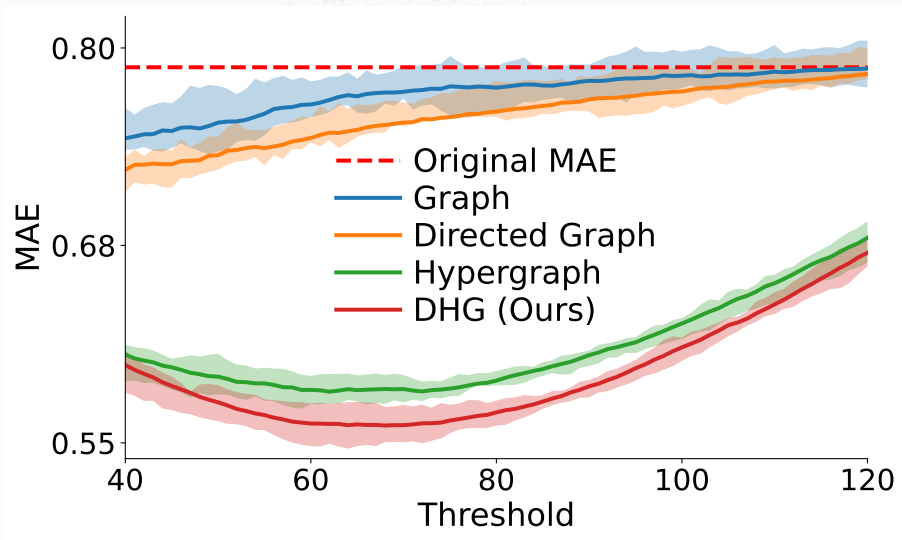
Baselines

- GSP — symmetric Laplacian
- DGSP — SVD of directed Laplacian
- t-HGSP — undirected tensor hypergraph

Metric: MAE vs. conserved frequencies

GSP-Traffic dataset [Liang et al., 2022]. Hyperarc construction: Zhao et al. (2024).

Denoising Results — Macheng, China



DHGSP (red) < GSP, DGSP, t-HGSP across all bandwidths. Shaded: 40th–60th percentile over 100 trials.

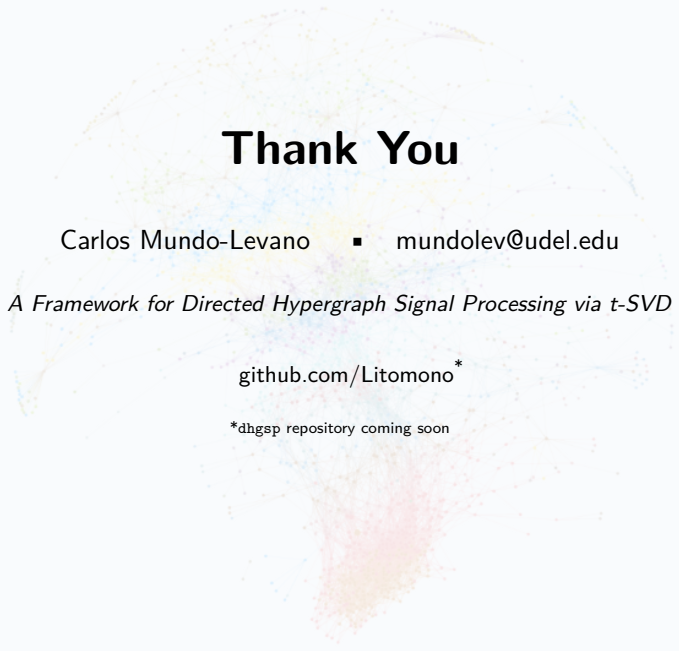
Conclusion

Contributions

- ① **B-hyperarc adjacency tensor** — canonical encoding, enables ②③
- ② **Lossless t-DHGFT** — self-adjoint dilation + t-SVD
- ③ **Traffic denoising** — outperforms GSP, DGSP, t-HGSP

Future Work

- Bandlimited sampling on directed hypergraphs
- Spectral clustering algorithms
- Directed hypergraph neural networks



Thank You

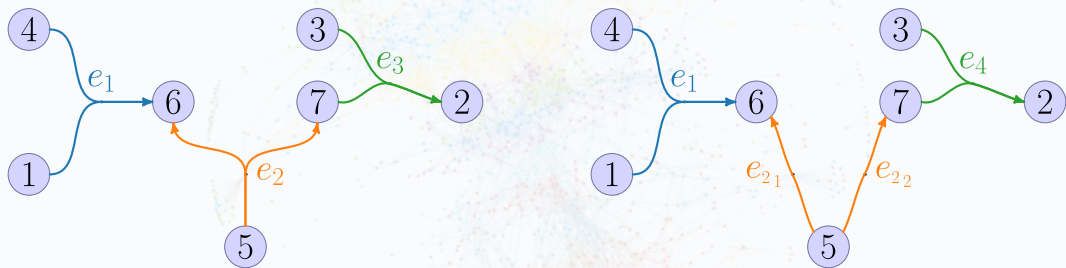
Carlos Mundo-Levano ■ mundolev@udel.edu

A Framework for Directed Hypergraph Signal Processing via t-SVD

github.com/Litomono*

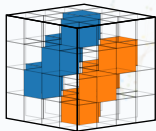
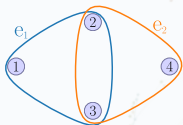
*dhgsp repository coming soon

B-Hyperarc Decomposition

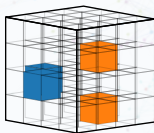


Asymmetry of $\mathcal{L}_{in}$ vs Undirected $\mathcal{L}$

Undirected Hypergraph

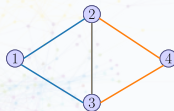


Directed Hypergraph



Frontal Slices are not symmetrical

Graph



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Directed Graph

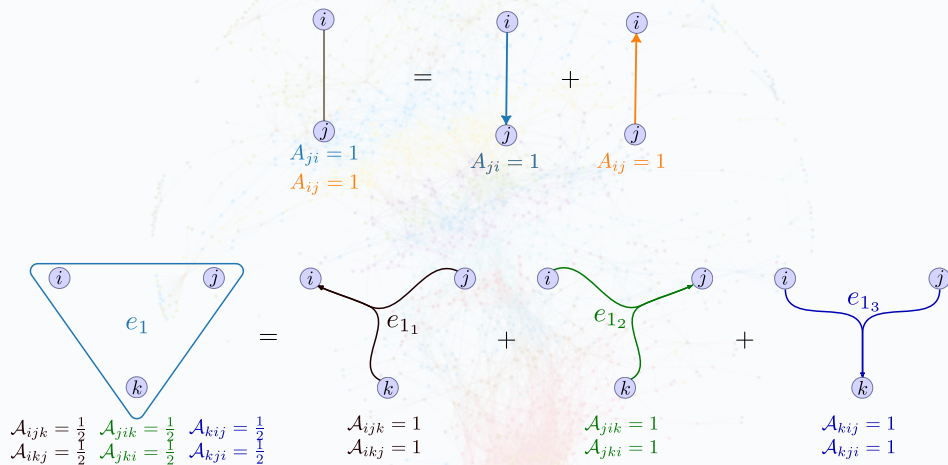


$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Matrix Representation
not symmetric

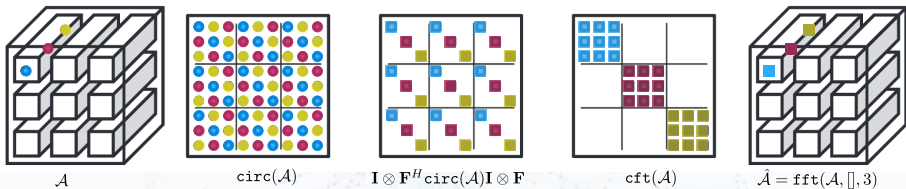
Full Eigen-Decomposition could
not exist $\iff$ no Lossless Fourier Transform
be no orthonormal $\iff$ no Parseval's Identity

Directed Hypergraph Subsumes Prior Structures



Gallo, L., et al. "Synchronization via directed higher-order interactions." Commun. Phys. (2022).

t-Product via DFT



$$\mathcal{C} = \mathcal{A} * \mathcal{B} := \text{ifft} \left(\left[\hat{\mathbf{A}}^{(k)} \hat{\mathbf{B}}^{(k)} \right]_{k=1}^{N_s}, [], 3 \right)$$

Kilmer, M., et al. "A third-order generalization of the matrix SVD." (2011).

t-SVD: Block Diagonalization via DFT

- Apply DFT along 3rd dim: $\hat{\mathcal{L}} = \text{fft}(\mathcal{L}, [], 3)$
- Each slice: $\hat{\mathbf{L}}^{(k)} = \hat{\mathbf{U}}^{(k)} \hat{\mathbf{S}}^{(k)} \hat{\mathbf{V}}^{(k)H}$ (matrix SVD)
- Apply IDFT: recover $\mathcal{U}, \mathcal{S}, \mathcal{V}$
- Cost: N_s independent $N \times N$ matrix SVDs

Kilmer, M., et al. "Factorization strategies for third-order tensors." Lin. Alg. Appl. (2011).

Adjacency Tensor: Normalization α_i

For hyperarc e_i with $c_i = |T(e_i)|$ distinct tail nodes, entries of length $M - 1$:

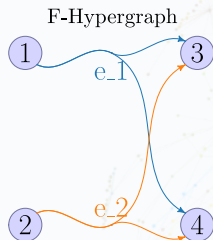
$$\alpha_i = \sum_{\substack{k_2, \dots, k_{c_i} \geq 1 \\ k_2 + \dots + k_{c_i} = M-1}} \frac{(M-1)!}{k_2! \cdots k_{c_i}!}$$

Example ($M = 3$):

$$|T| = 2: \alpha = \frac{2!}{1!1!} = 2 \quad \Rightarrow a = |T|/\alpha = 2/2 = 1$$

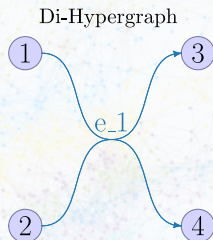
$$|T| = 1: \alpha = \frac{2!}{2!} = 1 \quad \Rightarrow a = 1/1 = 1$$

Shift Operator: Tensor Definitions Compared



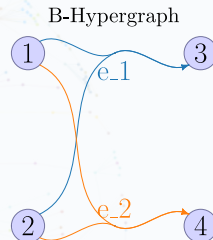
$$\vec{\mathcal{Y}}_F^{(0)} = \frac{1}{2} \begin{bmatrix} X_3 X_4 \\ X_3 X_4 \\ 0 \\ 0 \end{bmatrix}$$

$$\vec{\mathcal{Y}}_{F^t}^{(0)} = \frac{1}{4} \begin{bmatrix} 0 \\ 0 \\ X_4(X_1 + X_2) \\ X_3(X_1 + X_2) \end{bmatrix}$$



$$\vec{\mathcal{Y}}_{Di}^{(0)} = \frac{1}{3} \begin{bmatrix} X_2 X_3 X_4 \\ X_1 X_3 X_4 \\ 0 \\ 0 \end{bmatrix}$$

$$\vec{\mathcal{Y}}_{Di^t}^{(0)} = \frac{1}{3} \begin{bmatrix} 0 \\ 0 \\ X_1 X_2 X_4 \\ X_1 X_2 X_3 \end{bmatrix}$$



$$\vec{\mathcal{Y}}_B^{(0)} = \frac{1}{4} \begin{bmatrix} X_2(X_3 + X_4) \\ X_1(X_3 + X_4) \\ 0 \\ 0 \end{bmatrix}$$

$$\vec{\mathcal{Y}}_{B^t}^{(0)} = \frac{1}{2} \begin{bmatrix} 0 \\ 0 \\ X_1 X_2 \\ X_1 X_2 \end{bmatrix}$$